

PROJECT-BASED LEARNING

PHASE-II EVALUATION

Class Pilot

Your Smart Class Attendance Assistant

Team ID DSCPP-III-2026-T340
Domain Data Structures & C++
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Project Overview

Class Pilot is a menu-driven C++ application that helps a student record attendance and understand the likely effect of future absences. It combines current attendance with a weekly timetable, planned holidays and prediction tools.

- Records subjects, timetable entries, holidays and attendance.
- Calculates current attendance and estimates future outcomes.
- Helps identify subjects that need attention.
- Supports planning; it does not replace an institution's official record.

Problem Statement and Objectives

Problem

Knowing the current percentage is not always enough. A student also needs to know how missing a future class may affect the minimum attendance requirement.

Objectives

- Reduce repeated manual calculations.
- Provide subject-wise attendance visibility.
- Support what-if and recovery planning.
- Organise timetable, holidays and history in one console application.
- Demonstrate practical use of C++ and data structures.

Demonstration Data Used in This Presentation

Field	Demo value
Student	Rohan
Minimum attendance	75%
Semester	01/08/2026–30/11/2026 (illustrative)
Subjects	Mathematics, Physics, Computer Science, Chemistry, English
Attendance examples	Mathematics 18/20; Physics 16/20; Computer Science 19/22; Chemistry 15/20; English 17/20

These values are demonstration inputs for the presentation, not claims about an existing student's records. Use the date format requested by the running program.

Application Workflow

1. Configure the student, semester dates and minimum requirement.
2. Add the subjects or activities.
3. Add weekly timetable entries and planned holidays.
4. Record attendance for completed sessions.
5. Review the dashboard, history and risk ranking.
6. Use the what-if simulator, future prediction and recovery planner.
7. Save the current state and exit.

Main Menu: Complete Feature List

1. Configure semester and requirement
2. Add activity / subject
3. Add timetable entry
4. Add planned holiday
5. Record attendance
6. Attendance dashboard
7. Activity details
8. Daily reality check
9. What-if simulator
10. Future attendance prediction
11. Upcoming sessions
12. Attendance-risk priority
13. Sort activities by attendance
14. Attendance history
15. Search history by date
16. Search history by activity
17. Recovery planner
18. View timetable
19. View holidays
20. View activities
21. Database information
22. Project information
23. Save current state
24. Exit (0)

Program Launch and Menu

```
CLASS PILOT
Your Smart Class Attendance Assistant
A planner for attendance, future classes and what-if checks.

Semester setup is not completed yet.

MAIN MENU
 1. Configure semester and requirement
 2. Add activity / subject
 3. Add timetable entry
 4. Add planned holiday
 5. Record attendance
 6. Attendance dashboard
 ...
23. Save current state
 0. Exit
Enter your choice:
```

1. Configure Semester and Requirement

```
DEMO INPUT
Student name: Rohan
Semester start date: 01/08/2026
Semester end date:   30/11/2026
Minimum required attendance (%): 75

EXPECTED STATE SHOWN BY APPLICATION
Student: Rohan | Required: 75.00%
```

- Establishes the student's settings and date range.
- The application uses the minimum percentage for later checks.

2. Add Five Subjects / Activities

Code	Subject	Demo attendance
MATH101	Mathematics	18 / 20 = 90.00%
PHY101	Physics	16 / 20 = 80.00%
CSC101	Computer Science	19 / 22 = 86.36%
CHEM101	Chemistry	15 / 20 = 75.00%
ENG101	English	17 / 20 = 85.00%

Purpose

Each subject is stored as an activity so timetable entries and attendance records can be associated with it.

3. Add Timetable Entries

```
ILLUSTRATIVE WEEKLY TIMETABLE
MON 09:00-10:00  MATH101  Mathematics
MON 10:15-11:15  PHY101   Physics
TUE 09:00-10:00  CSC101   Computer Science
WED 11:00-12:00  CHEM101  Chemistry
THU 09:00-10:00  ENG101   English
FRI 10:15-11:15  CSC101   Computer Science
```

- Weekly entries are used to identify future sessions.
- The timetable can be reviewed using menu option 18.

4. Add Planned Holiday

```
DEMO HOLIDAY
Date: 15/08/2026
Description: Independence Day

EXPECTED EFFECT
The date is treated as a planned holiday when upcoming sessions
are generated for the configured semester.
```

- Holidays help avoid counting a non-teaching day as a regular class.
- View the holiday list with menu option 19.

5. Record Attendance

```
DEMO ATTENDANCE ENTRIES
Date: 03/08/2026 | Activity: MATH101 | Present
Date: 03/08/2026 | Activity: PHY101  | Present
Date: 04/08/2026 | Activity: CSC101  | Absent
Date: 05/08/2026 | Activity: CHEM101 | Present
Date: 06/08/2026 | Activity: ENG101  | Present
```

- Each record connects an activity, date and attendance status.
- Records feed the dashboard, history and prediction tools.

6. Attendance Dashboard

```
ATTENDANCE DASHBOARD -- DEMONSTRATION
Student: Rohan | Required: 75.00%

Subject              Attended  Conducted  Attendance
Mathematics              18        20        90.00%
Physics                  16        20        80.00%
Computer Science         19        22        86.36%
Chemistry                15        20        75.00%
English                  17        20        85.00%
```

7. Activity Details

```
ACTIVITY DETAILS -- CHEM101
Name: Chemistry
Attended: 15
Conducted: 20
Current attendance: 75.00%
Minimum required: 75.00%
Status: AT THE MINIMUM REQUIREMENT
```

- Gives a focused view of one subject.
- At exactly 75%, any future absence may reduce the percentage below the threshold.

8. Daily Reality Check

```
DAILY REALITY CHECK -- DEMO SCENARIO
Date: 10/08/2026
Student: Rohan | Required: 75.00%
Scheduled: Computer Science, 09:00-10:00
Current attendance: 86.36%
Safe absences currently available: 3

This is only a planning estimate.
The official institutional record remains final.
```

9. What-if Simulator

```
WHAT-IF EXAMPLE -- CHEMISTRY
Current attendance: 15 / 20 = 75.00%
Scenario: miss the next 1 class
Projected attendance: 15 / 21 = 71.43%
Result: the plan goes below the requirement.
Estimated recovery classes: 3
```

Interpretation

The simulation helps Rohan understand the effect of a proposed absence before making a decision.

10. Future Attendance Prediction

```
FUTURE PREDICTION -- DEMO
Activity: MATH101
Current attendance: 18 / 20 = 90.00%
Upcoming sessions in selected range: 8
Scenario: miss 2 sessions
Projected attendance: 18 / 22 = 81.82%
Result: ABOVE MINIMUM REQUIREMENT
```

- Uses the configured timetable and selected date range.
- Planned holidays are excluded when sessions are generated.

11. Upcoming Sessions

```
UPCOMING SESSIONS -- ILLUSTRATIVE
The queue contains 4 upcoming sessions:

10/08/2026  09:00-10:00  CSC101  Computer Science
11/08/2026  09:00-10:00  CSC101  Computer Science
12/08/2026  11:00-12:00  CHEM101 Chemistry
13/08/2026  09:00-10:00  ENG101  English
```

- Future sessions are generated from the timetable and semester dates.
- A queue is used to process the next sessions in order.

12. Attendance-risk Priority

```
ATTENDANCE-RISK PRIORITY -- DEMO
Highest-risk activity shown first:

Chemistry (CHEM101)
  Attendance: 75.00%
  Can safely miss: 0

Physics (PHY101)
  Attendance: 80.00%
  Can safely miss: 1
```

- Prioritises subjects with less attendance flexibility.
- A priority queue can return the highest-risk item first.

13. Sort Activities by Attendance

```
SORTED ATTENDANCE -- LOWEST TO HIGHEST (DEMO)
Chemistry          75.00%
Physics            80.00%
English            85.00%
Computer Science   86.36%
Mathematics        90.00%
```

- Sorting makes the subjects requiring attention easier to identify.
- The source uses sorting with a comparator for ordered results.

14. Attendance History

```
ATTENDANCE HISTORY -- DEMO
Date        Activity   Status
03/08/2026  MATH101    Present
03/08/2026  PHY101     Present
04/08/2026  CSC101     Absent
05/08/2026  CHEM101    Present
06/08/2026  ENG101     Present
```

- Displays recorded sessions for review.
- Provides the underlying records used by summaries and searches.

15. Search History by Date

```
SEARCH HISTORY BY DATE -- DEMO
Search date: 04/08/2026

Matching record:
04/08/2026  09:00-10:00  CSC101  Absent
```

- Helps locate all attendance entries for a particular date.
- If no record matches, the program reports that no record was found.

16. Search History by Activity

```
SEARCH HISTORY BY ACTIVITY -- DEMO
Activity code: CHEM101

05/08/2026  Chemistry  Present
07/08/2026  Chemistry  Present
10/08/2026  Chemistry  Absent
```

- Filters history for one selected subject/activity.
- Useful for tracing how a subject's attendance changed over time.

17. Recovery Planner

```
RECOVERY PLANNER -- DEMO
Activity: Chemistry
Current attendance: 15 / 21 = 71.43%
Minimum required: 75.00%

Classes that need to be attended consecutively to reach
75.00%: 3
This assumes every one of those future classes is attended.
```

- Recovery planning is relevant when attendance is below the threshold.
- From $15/21$, attending 3 consecutive classes gives $18/24 = 75.00\%$.

18. View Timetable

```
WEEKLY TIMETABLE -- DEMO
MON 09:00-10:00  MATH101  Mathematics
MON 10:15-11:15  PHY101   Physics
TUE 09:00-10:00  CSC101   Computer Science
WED 11:00-12:00  CHEM101  Chemistry
THU 09:00-10:00  ENG101   English
FRI 10:15-11:15  CSC101   Computer Science
```

19. View Holidays

```
PLANNED HOLIDAYS -- DEMO
15/08/2026  Independence Day
```

- Provides a quick review of dates marked as holidays.
- Holiday entries influence future-session generation.

20. View Activities

```
ACTIVITIES -- DEMO
MATH101  Mathematics
PHY101   Physics
CSC101   Computer Science
CHEM101  Chemistry
ENG101   English
```

- Lists the subjects currently registered in the application.
- Activity codes connect the subject to timetable and attendance data.

21. Database / Storage Information

```
STORAGE INFORMATION
The application uses a standard C++ text file:
attendance_data.txt

Information categories represented in the application:
- Student settings
- Activities / subjects
- Timetable entries
- Planned holidays
- Attendance records
```

Environment note

The source version used for this deck uses file-based storage, not SQLite. A restricted online compiler may prevent writing files; if so, persistence cannot be demonstrated in that environment.

22. Project Information

Class Pilot

A C++ attendance-planning application that helps answer: “If I miss a future session, what may happen to my attendance?”

Data structures and algorithms used

- `vector`: stores collections of activities, timetable entries, holidays and history.
- `unordered_map`: supports activity lookup by code.
- `queue`: processes upcoming sessions.
- `priority_queue`: ranks attendance risk.
- `sort`: orders attendance results.

23. Save Current State

```
SAVE CURRENT STATE
Current important records are saved in attendance_data.txt.
```

- The program attempts to save the current settings and records.
- Successful persistence depends on write permission in the execution environment.
- In a restricted online sandbox, file saving may fail; this should be reported honestly in a live demo.

Attendance Calculation

$$\text{Attendance \%} = \frac{\text{Classes attended}}{\text{Classes conducted}} \times 100$$

$$\text{Example} = \frac{18}{20} \times 100 = 90\%$$

Safe future absences

The application tests how the percentage changes when future classes are added to the total without increasing the attended count.

Recovery

If attendance is below the requirement, it estimates how many consecutive future classes must be attended to reach the threshold.

Testing and Demonstration Checklist

No.	Feature	Demonstration check
1–5	Setup, activities, timetable, holidays, attendance	Enter demo data and inspect prompts
6–8	Dashboard, details, daily check	Compare displayed totals and requirement
9–12	What-if, prediction, upcoming, risk	Verify date range and scenario assumptions
13–17	Sort, history, searches, recovery	Check ordering, filters and arithmetic
18–20	Timetable, holidays, activities	Confirm listed entries match setup
21–23	Storage info, project info, save	Check messages and write permissions
0	Exit	Confirm clean exit message

Limitations and Responsible Use

- Results depend on the correctness of the information entered by the user.
- Timetable changes must be updated manually.
- A restricted online compiler may not retain data files between runs.
- Predictions are estimates and do not override official attendance records or institutional rules.

Conclusion

Class Pilot brings attendance recording and attendance planning together in a single C++ console application. Its features allow a student to review current attendance, explore possible future outcomes, identify subjects at risk and estimate recovery needs. The project also demonstrates how common data structures can support practical application features.

Key takeaway

Class Pilot helps the student make a more informed attendance decision; it does not make the decision for them.

Thank You

Class Pilot

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